

ACCELERATING LEARNED SAR IMAGE COMPRESSION VIA SELECTIVE CHANNEL-GROUP ENCODING BYPASS

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ABSTRACT

Recent modifications to the Efficient Learned Image Compression (ELIC) framework have shown state-of-the-art rate-distortion performance for synthetic aperture radar (SAR) image compression. However, the entropy coding stage can be computationally expensive with minimal compression gains when the images' latent features contain a high proportion of repetitive values, which is prevalent in SAR data preprocessed via Discrete Cosine Transform (DCT). We propose a simple method for accelerating SAR compression with a configurable bypass of ELIC's asymmetric numeral systems (ANS) encoding in favor of traditional run-length encoding (RLE) for specific latent channel groups. This approach provides a level of control over the bitrate-speed trade-off with no reduction in image reconstruction quality. Experiments on 256×256 patches from the National Geospatial-Intelligence Agency (NGA) SAR dataset show that partial bypass configurations can achieve anywhere from 17% runtime savings at 2% bitrate cost to 83% runtime savings at 50% bitrate cost. Our analysis highlights how different latent feature characteristics and encoding algorithms contribute to the system's bitrate and computational complexity, offering insights for future learned image compression model designs.

Index Terms— Synthetic Aperture Radar (SAR), Learned Image Compression, Run-Length Encoding, DCT

1. INTRODUCTION

SAR is an active imaging sensor used extensively in environmental monitoring, agriculture, and military reconnaissance. Unlike passive optical sensors, SAR transmits microwave pulses toward Earth's surface and reconstructs images from the amplitude and phase of the backscattered signals. The sensor is typically installed on a satellite or other aerial platform, whose motion over the surface allows data to be collected across a wide area and combined via sophisticated signal processing algorithms. This process simulates a much larger antenna, and the result is a single high-resolution image. SAR imagery is also robust to many environments where optical sensors fail, including dark, cloudy, and all-weather

conditions. The amplitude (A) and phase (ϕ) of the returned signal are usually represented as in-phase (I) and quadrature (Q) components and stored as a single complex value ($I + jQ$). These representations can be converted using the following relationship:

$$A = \sqrt{I^2 + Q^2} \quad (1)$$

$$\phi = \arctan(Q/I) \quad (2)$$

SAR images are typically stored in 32-bit floating point precision per component per pixel, with spatial resolutions often exceeding 1000×1000 , which results in file sizes reaching hundreds of megabytes. This creates significant storage and transmission challenges, as satellites and other airborne platforms have limited local storage and downlink bandwidth. The images are often compressed via traditional or learned methods, aiming for an ideal balance of compression ratio and image fidelity, known as the rate-distortion trade-off. However, onboard compression must also be lightweight, due to real-time processing constraints and limited computational resources. Thus, the objective becomes a three-way trade-off between bitrate, distortion, and complexity.

Conventional compression methods offer low computational complexity, but they are limited in their ability to adapt to spatial dependencies and account for the speckle noise prevalent in coherent imaging systems like SAR. Deep-learning-based compression frameworks achieve state-of-the-art rate-distortion performance on optical images by exploiting spatial redundancies, but they are often less effective on SAR imagery, and their learned entropy coding models can be unnecessarily complex on highly redundant data streams. Quantized latent representations exhibiting highly-skewed distributions, low energy, or runs of repeated values may be better suited by a simpler, static coding system that avoids the computational overhead of adaptive context modeling.

In this work, we investigate the rate-complexity trade-off in learned compression of complex-valued SAR images. Specifically, we analyze several variations of the entropy coding stage to better understand how compression efficiency and computational cost can be balanced for resource-limited SAR systems. Our main contributions are the following:

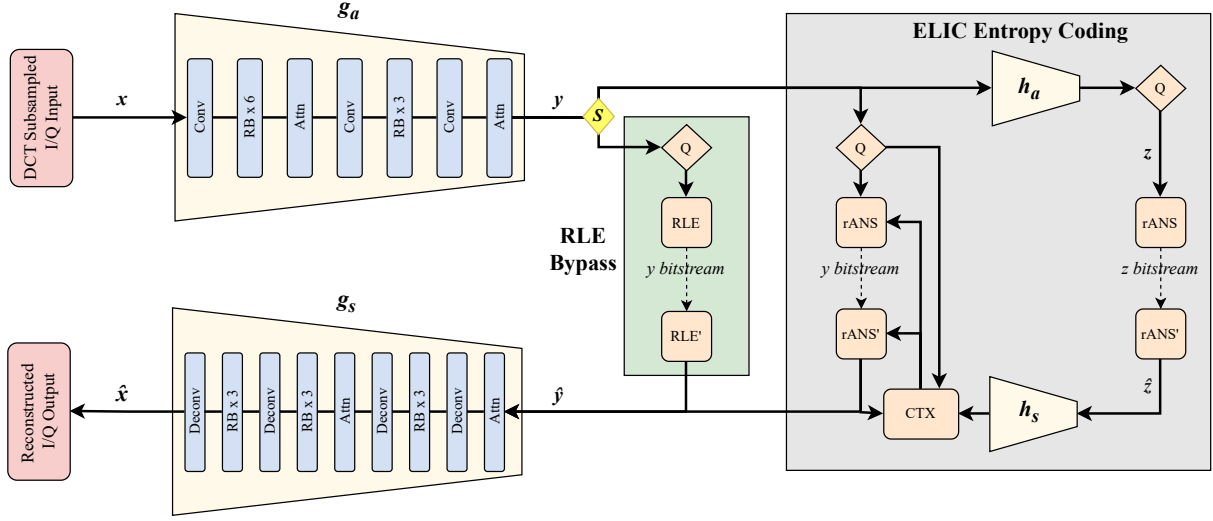


Fig. 1. Diagram of the ELIC architecture with our proposed entropy coding bypass. The switch S routes each latent channel group to its pre-determined encoding module, either RLE Bypass or ELIC Entropy Coding. The channel groups' assignments are configured prior to model training and are fixed during inference.

- We introduce a configurable approach for accelerating learned SAR compression by selectively bypassing ELIC's learned entropy coding stage for certain latent channels in favor of traditional run-length encoding (RLE).
- Through systematic testing, we evaluate multiple bypass configurations to characterize the resulting trade-offs between bitrate, distortion, and encoding/decoding time.

2. RELATED WORK

SAR image compression presents unique challenges due to the complex-valued nature of the data, with correlated but distinct amplitude and phase components that must be preserved for downstream applications. Traditional compression approaches like JPEG [1], JPEG2000 [2], HEVC [3], and VVC [4] rely on transform-based methods such as DCT and wavelet coding, often combined with entropy coding techniques like Huffman coding or adaptive arithmetic coding. These methods are typically fast and reliable, but they are ill-equipped to handle the high prevalence of speckle noise in SAR imagery, often resulting in visible artifacts or distortion [5–9].

In recent years, learned image compression (LIC) methods have demonstrated strong rate–distortion performance, particularly the variational hyperprior framework of Ballé *et al.* [10]. This was later refined in the ELIC architecture [11], whose space-channel context model employs parallel checkerboard convolution [12] for accelerated decoding.

Maharjan *et al.* [13] extended this paradigm to complex-valued SAR data, introducing an end-to-end learned compression framework that serves as the foundation for this work.

Beyond rate and distortion, recent studies have emphasized computational complexity as a third axis of optimization [14–19]. Inspired by these trends, we explore entropy coding variations that improve encoding efficiency while maintaining competitive rate–distortion performance.

3. PROPOSED METHOD

3.1. Network Architecture

Our approach builds upon the learned SAR compression framework proposed by [13]. Due to the band-limited nature of SAR imagery, most valuable information is contained in the low-frequency terms, which can be exploited by applying a 2-dimensional 4×4 Discrete Cosine Transform (DCT) and subsampling the resulting frequencies. This results in a reduced spatial resolution with no information loss. Next, the DCT subband image is input to the ELIC architecture for compression, which follows a standard variational autoencoder structure consisting of an analysis network, entropy model, and synthesis network. The analysis network encodes the input image into a 320-channel latent representation, which is then quantized using uniform scalar quantization. A hyperprior network provides side information for entropy modeling, while a checkerboard spatial context [12] and autoregressive channel-group context [20] refine likelihood estimation by exploiting spatial and inter-channel dependen-

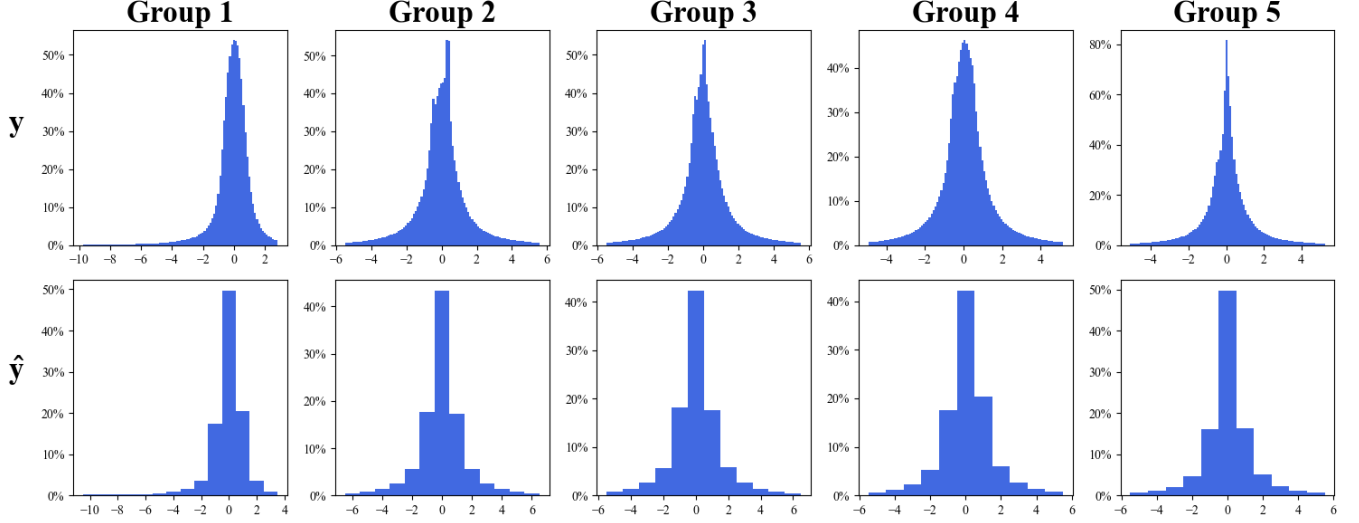


Fig. 2. Distributions of the unquantized (y) and quantized ($\hat{y}$) latent features within each channel group.

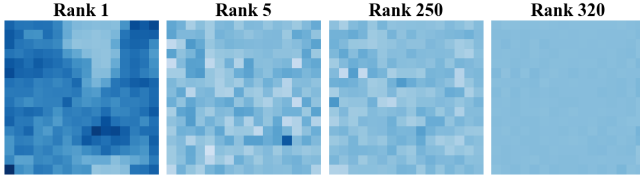


Fig. 3. Visualization of latent feature values for a single test image, showing a variety of channel energy. Displayed here are Channel 1 ($var = 5.3$), Channel 155 ($var = 2.1$), Channel 78 ($var = 0.9$), and Channel 272 ($var = 0.02$).

cies. Finally, the range variant of the asymmetric numeral systems method (rANS) is used for entropy coding. During decoding, a synthesizer network directly reconstructs the I/Q image in the pixel domain, eliminating the need for an inverse DCT step.

The end-to-end model is trained using a traditional Rate-Distortion Optimization (RDO) scheme with the following loss function:

$$L = R(\hat{y}) + \lambda D(x, \hat{x}) \quad (3)$$

The rate term R represents the measured bitrate (bits per pixel) of the encoded signal, representing the realized entropy of the quantized latent features. D denotes the distortion (mean squared error) between the original and reconstructed images. λ is a Lagrange multiplier that can be varied to generate multiple operating points along a rate-distortion curve.

3.2. Selective Encoding Bypass

In this work, we retain the same architecture and pre-trained weights for a single model from [13], trained with $\lambda = 1$.

However, specific latent channel groups are pre-selected to bypass the learned entropy coding stage. For bypassed groups, the quantized latents are encoded using simple run-length and Huffman coding, while the remaining groups are encoded using the standard ELIC entropy model. The selection of channel groups is fixed for each experiment and does not depend on image content. Our analysis evaluates multiple such configurations to characterize their impact on bitrate and encoding/decoding time relative to the baseline codec with no bypass. Since RLE and Huffman coding are fully invertible, the bypass has no meaningful impact on image reconstruction quality. The full architecture is shown in Fig. 1.

3.3. Dataset and Experimental Setup

The model used for this work was trained on the NGA SAR dataset. The original dataset consists of two high-resolution images of 3800×6600 pixels each, from which 10,000 patches of size 256×256 were extracted for training. An additional 1,000 patches of the same size were used for evaluation. Each patch was pre-processed using 2-dimensional 4×4 DCT, resulting in a 64×64 input to the model.

Ten total bypass configurations were tested. The baseline configuration used an unmodified version of the model with no channel group bypass. Next, each of the five channel groups was individually bypassed. The final four configurations were obtained by bypassing Groups 1-2, 1-3, 1-4, and 1-5. To estimate computational complexity, we measured the time required for latent feature encoding and decoding. For the existing ELIC entropy coding procedure, this included the generation of the hyperprior, quantization, and rANS-based encoding/decoding. For the RLE bypass, this included only the quantization and RLE encoding/decoding of the latent features. Each experiment consisted of one warm-up iteration

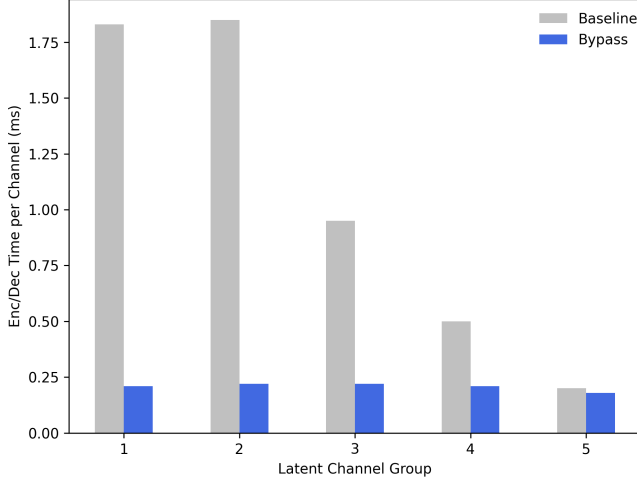


Fig. 4. Comparison of combined encoding/decoding time per channel for the ELIC entropy coding baseline and the RLE bypass. The baseline complexity is heavily dependent on channel energy, while the bypass complexity is not.

and ten timed iterations to compute the mean.

All timed tests were performed back-to-back on the same system, equipped with two AMD EPYC 9554 CPUs (128 cores @ 3.76 GHz), 810 GB RAM, and one Nvidia L40S GPU (51.5 GB VRAM). The environment used Rocky Linux 8.10, Python 3.10, and PyTorch 2.5.1 (with CUDA 12.4 build, 12.6 runtime).

3.4. Performance Metrics

Bitrate is reported here as bits per pixel per channel, where $F_{\hat{y}}$ represents the compressed file size in bytes, and each test image contains $C = 2$ channels and a spatial resolution of $(H, W) = (256, 256)$:

$$BPP = \frac{8 \times F_{\hat{y}}}{C \times H \times W} \quad (4)$$

While the primary focus of our analysis is compression rate versus computational cost, we also track mean Signal-to-Quantization-Noise Ratio (SQNR) to evaluate the impact on reconstruction quality of the SAR amplitude image, using a neighborhood size of $k = 5$.

4. RESULTS

4.1. Latent Channel Analysis

The primary purpose of this work is to characterize the rate-complexity trade-off introduced by selectively bypassing ELIC’s learned entropy coding framework. Complexity is reduced by exploiting redundancies in a latent channel’s signal, which can be represented quickly and efficiently by RLE. The model’s 320 latent channels are first collected into five

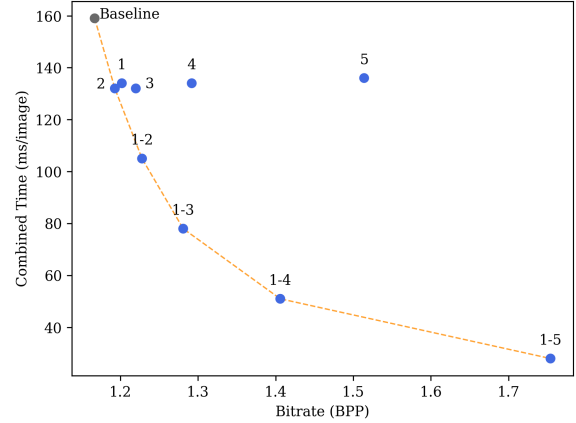


Fig. 5. Visualization of bitrate versus combined encoding/decoding time for each configuration tested. The dotted line represents the Pareto frontier of these variants, resulting in a rate-complexity curve that can benefit future studies.

groups of sizes 16, 16, 32, 64, and 192, without modifying the channel order. The distribution of each latent channel group before and after quantization is shown in Fig. 2. Most latent values cluster near zero, resulting in a sparsity pattern well-suited for a variable-length coding (VLC) scheme. Fig. 3 shows four example latent channels from a single test image, ranked from highest to lowest energy.

We empirically find that earlier latent channels contain more energy than later channels, on average. As shown in Fig. 4, this decrease in energy across channels corresponds to a decrease in encoding and decoding times for ELIC’s entropy model. For example, the original ELIC architecture requires 1.83 ms per channel in Group 1 but only 0.20 ms per channel in Group 5. In contrast, the RLE coding scheme is relatively invariant with respect to channel energy, resulting in roughly uniform encoding/decoding times per channel across all five groups.

4.2. Rate-Complexity Trade-Off

Our experimental results highlight an inherent trade-off between compression efficiency and computational cost when selecting the encoding method for each channel group. Fig. 5 shows that the various configurations define multiple useful operating points along this spectrum.

Among individual group bypasses, only the Group 2 variant lies on the Pareto frontier, dominating the performance of Groups 1, 3, 4, and 5. Together with the Group 2 bypass, the multi-group bypass configurations produce a rate-complexity curve, ranging from optimal bitrate to minimal computational cost. For example, the Group 2 bypass achieves roughly 17% lower complexity at a modest 2% bitrate increase, while the

Table 1. Performance comparison for RLE bypass variants.

Variant	# Bypassed	BPP	SQNR	Enc (ms)	Dec (ms)	Total (ms)	Δ BPP (%)	Δ Time (%)
Baseline	0	1.167	1.529	102	57	159	–	–
Grp 1	16	1.202	1.530	85	49	134	3.0	–15.9
Grp 2	16	1.193	1.529	84	48	132	2.2	–17.3
Grp 3	32	1.220	1.530	84	48	132	4.6	–17.1
Grp 4	64	1.292	1.532	84	50	134	10.7	–16.0
Grp 5	192	1.514	1.534	82	54	136	29.8	–14.7
Grps 1-2	32	1.228	1.530	66	39	105	5.2	–34.4
Grps 1-3	64	1.281	1.531	48	30	78	9.8	–51.2
Grps 1-4	128	1.406	1.534	29	22	51	20.5	–67.8
Grps 1-5	320	1.754	1.538	9	18	28	50.3	–82.6

Group 1-3 bypass achieves 51% lower complexity with a 10% bitrate increase. The full results for each configuration are shown in Table 1. "Grp 1" refers to the variant which bypassed only channel group 1, and "Grps 1-2" refers to the variant which bypassed groups 1 through 2. The total number of channels bypassed for each variant is shown in the "# Bypassed" column.

Importantly, due to the shared ELIC analysis transform and lossless coding procedure, the RLE bypass does not meaningfully impact reconstruction quality. All tested variations produced marginally higher SQNR than the baseline, as a result of small differences in quantization. These results demonstrate that selective entropy coding bypasses can flexibly adjust computational cost while maintaining acceptable compression efficiency, allowing end users to choose the most appropriate balance for their application.

5. CONCLUSION

The rate-complexity curve generated by our selective RLE bypass approach allows for a degree of control over computational cost, and future experiments may reveal additional viable configurations along the Pareto frontier. The DCT-ELIC model used here is trained specifically for rANS coding using a space-channel context model. A model that incorporates a lightweight conditional entropy model and is trained end-to-end for these hybrid coding configurations will likely achieve better performance. While these future refinements may further improve coding efficiency, our proposed approach delivers a practical set of configurations which reduce computational cost with minimal bitrate impact and simple implementation, enabling faster and more cost-effective SAR storage and transmission in resource-limited applications.

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